The Invariance Principle — problem 1.27

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Three automata I, R, S print pairs of positive integers on tickets. For entry (x, y), the automata I, R, S give tickets $(x-y, y)$, $(x+y, y)$, (y, x), respectively, as outputs. Initially, we have the ticket $(1, 2)$. With these automata, can I get the tickets (a) $(19, 79)$ (b) $(819, 357)$? Find an invariant. What pairs (p, q) can I get starting with (a, b) ? Via which pair should I best go ?